SPLITTING MADSEN-TILLMANN SPECTRA VIA TWISTED TRANSFER MAPS

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1. Introduction

In [GMTW09], Galatius, Madsen, Tillmann, and Weiss defined for each nonnegative integer $d \geq 0$, a spectrum $MT\theta(d)$, whose infinite loop space deloops to the classifying space $B\mathsf{Cob}(d)_\theta$ of a particular topological category. The category $\mathsf{Cob}(d)_\theta$ parametrizes cobordisms between $(d-1)$-dimensional closed smooth manifolds with θ-structures for a given tangential structure $\theta : B \to BO(d)$. The homotopy invariants of these spectra, called *Madsen-Tillmann spectra*, contain information about diffeomorphism groups of manifolds [GRW10], as well as applications to physics via topological quantum field theories [FH21]. These applications have sparked great interest in using techniques from stable homotopy theory to better understand and classify Madsen-Tillmann spectra.

One of the most effective means of studying Madsen-Tillman spectra came in the work of Kashiwabara and Zare [KZ18], which made use of the transfer maps of [BG74] and [MMM86] to give splittings of the p-localized spectrum $MT\boldsymbol{K}(n)_{(p)}$ for $\boldsymbol{K} = O, SO, U, SU, Sp$ and certain primes p.

For example, Kashiwabara and Zare were able to give a complete characterization of the p-local structure of the Madsen-Tillman spectra for manifolds with no tangential structure at an odd prime p.

Theorem 1.1. *Given p an odd prime and $n \geq 0$, we have a homotopy equivalence*

$$MTO(n)_{(p)} \simeq \begin{cases} \Sigma^\infty_+ BO(2n)_{(p)}, & n \text{ even}, \\ *, & n \text{ odd}. \end{cases}$$

This expository work aims to develop the splitting results of [KZ18], in the cases $\boldsymbol{K} = O$ and SO, at a level accessible for a student that has completed a standard graduate sequence of algebraic and differential topology.

We fix some notation that will be used throughout.

Notations 1.2.

1. We denote arbitrary Lie groups by G, K, H and their associated Lie algebras by $\mathfrak{g}, \mathfrak{k}, \mathfrak{h}$.
2. We use boldface math $\boldsymbol{G}, \boldsymbol{K}$ to denote the symbols O, SO. Hence if $\boldsymbol{K} = O$, $\boldsymbol{K}(n)$ denotes the real Lie group of orthogonal matrices of size n.
3. Given a vector bundle γ over a space X and a continuous map $f : Y \to X$, we denote the pullback $f^*\gamma$ by $\gamma|_Y$ when f is clear from context.
4. We denote by $-\gamma$ the associated virtual vector bundle in the Grothendieck group $K^0(X)$ associated to the monoid $\mathsf{Vect}_{\mathbf{k}}(X)$ of isomorphism classes of finite-dimensional $\mathbf{k}$-vector bundles over X with addition given by Whitney sum.

1.1. A brief word on classifying spaces and principal G-bundles.

Throughout the paper, we make use of several standard results concerning principal G-bundles and vector bundles. Those which will be used repeatedly or which we believe may not be familiar to many readers are summarized below. All of the results can be found in [Mit17].

Definition 1.3. A subgroup H of a topological group G is **admissible** if the quotient map $G \to G/H$ is a principal H-bundle.

The main instance of admissible subgroups we will be interested in is the following.

Proposition 1.4. *A closed subgroup K of a Lie group G is admissible.*

One of the main applications of admissible subgroups is the following.

Proposition 1.5. *Suppose $K \leq G$ is an admissible subgroup. Letting $i : K \hookrightarrow G$ denote the inclusion homomorphism, we have a weak homotopy equivalence $BK \simeq EG/K \cong EG \times_G G/K$, and furthermore under this model, the map $Bi : BK \to BG$ is given by the projection of the fiber bundle associated to EG with fiber G/K.*

The following fact will be used repeatedly throughout our proofs.

Proposition 1.6. *Let E be a principal G-space over B and F a (left) G-space. Given a continuous map $f : B' \to B$, we have a natural isomorphism of fiber bundles with structure group G:*

$$f^*(E \times_G F) \cong f^*(E) \times_G F.$$

1.1.1. *Classifying spaces.* Given $\boldsymbol{K} = O, SO$ and $n \geq 0$, we let $B\boldsymbol{K}(n)$ denote the classifying space for n-dimensional real vector bundles with $\boldsymbol{K}(n)$-structure. Recall, given a CW-complex B, a real vector bundle V of rank n over B is classified by a map $\xi : B \to BGL_n(\boldsymbol{R})$, and a $\boldsymbol{K}(n)$*-structure* on V, for $\boldsymbol{K}(n) \subset GL_n(\mathbf{R})$, amounts to a reduction of the structure group of V to $\boldsymbol{K}(n)$, or equivalently, the existence of a dashed lift making the following diagram commute up to homotopy:

$$\begin{tikzcd} & \boldsymbol{K}(n) \arrow[d, hook] \\ B \arrow[ur, dashed, "\widetilde{\xi}"] \arrow[r, "\xi"'] & BGL_n(\mathbf{R}) \end{tikzcd}.$$

The Gram-Schmidt process defines a homotopy equivalence $BGL_n(\mathbf{R}) \simeq BO(n)$, which corresponds to the geometric fact that vector bundles over a paracompact space can naturally be equipped with a Riemannian metric. Thus we can naturally consider rank n vector bundles as being classified by maps into $BO(n)$.

Warning 1.7 (The case $n = 0$). Note by our above definition, it is *not* necessarily the case that $B\boldsymbol{K}(n)$ denotes the classifying space of $\boldsymbol{K}(n)$. In particular, $BSO(0) \simeq S^0$ (encoding the fact that a 0-dimensional real vector space admits two orientations), while the classifying space of $SO(0) = 0$ is contractible.

Given $n \geq 1$ and $\boldsymbol{K} = O, SO$, we fix a well-known model for the classifying space $B\boldsymbol{K}(n)$ and the corresponding tautological n-plane bundle $\gamma_n^{\boldsymbol{K}}$ over $B\boldsymbol{K}(n)$, classifying vector bundles with $\boldsymbol{K}(n)$-structure.

1. We define $BO(n) \coloneqq \mathrm{Gr}_n(\mathbf{R}^\infty)$, the infinite Grassmannian of n-planes in $\mathbf{R}^\infty$. The total space U_n of γ_n^O is defined so that the fiber over the n-plane $V \in BO(n)$ is itself, topologized as a subspace of the product $BO(n) \times \mathbf{R}^\infty$ and equipped with the obvious projection map.
2. $BU(n) \coloneqq \mathrm{Gr}_n^+(\mathbf{R}^\infty)$, the infinite Grassmannian of oriented n-planes in $\mathbf{R}^\infty$. The tautological n-plane bundles γ_n^{SO} is defined analogously to the unoriented case.
3. We define $E\boldsymbol{K}(n) \coloneqq \mathrm{Fr}^{\boldsymbol{K}}(U_{\boldsymbol{K}})$, the frame bundle associated to the tautological bundle $\gamma_n^{\boldsymbol{K}}$.

1.2. **Representation theory.**

We give a recollection of some basic representation theory for matrix Lie groups, which will be necessary to construct the transfer maps of spectra we use throughout.

Given a Lie group G, the Lie algebra $\mathfrak{g}$ is defined as the tangent space T_eG of G at the identity $e \in G$. In the case of matrix Lie groups, i.e., a Lie subgroup of $M_n(\mathbf{C})$ for some $n \geq 1$, the associated Lie algebras have particular straightforward.

Lemma 1.8. *The Lie algebra of $O(n)$ (and hence $SO(n)$ as the connected component of the identity) can be identified with the real subalgebra of $M_n(\mathbf{R})$ of skew-symmetric matrices:*

$$\mathfrak{o}(n) = \mathfrak{so}(n) = \{A \in M_n(\mathbf{R}) : A^* = -A\}.$$

The Lie algebra $\mathfrak{g}$ admits a G-representation, called the *adjoint representation*, defined as follows. Let $\Psi : G \to \mathrm{Aut}(G)$ by the Lie group homomorphism defined by taking $g \in G$ to the inner automorphism $\Psi(g) : x \mapsto gxg^{-1}$. Taking the differential at the identity $e \in G$ of an inner automorphism $\Psi(g)$ defines a Lie algebra endomorphism $\mathrm{Ad}_G(g) \coloneqq d_e\Psi(g) \in \mathrm{Lie}(\mathfrak{g})$. A routine chain rule computation implies the assignment $g \mapsto \mathrm{Ad}_G(g)$ defines a group homomorphism

$$\mathrm{Ad}_G : G \to \mathrm{Lie}(\mathfrak{g}).$$

Definition 1.9. The representation Ad_G is called the **adjoint representation** of the Lie group G.

In the case when G is a matrix Lie group, the adjoint representation of G has a particularly straightforward description in terms of conjugation by G.

1.3. **Thom spectra.**

We denote by $\mathrm{Ho}(\mathsf{Sp})$ the stable homotopy category of Bousfield and Friedlander. It is given by the homotopy category of a symmetric monoidal model category of spectra $(\mathsf{Sp}, \wedge, \mathbf{S}^0)$, of which there are several (Quillen-)equivalent models [MMSS01]. Since we are only interested in constructions within the stable homotopy category, for simplicity we implicitly work in the category of sequential spectra, where the weak equivalences are given by π_*-isomorphisms, cofibrations are given levelwise via the Quillen model structure on Top_*, and fibrations are defined via a suitable right lifting property.

Definition 1.10. Given a vector bundle V on a space X, we define the **Thom spectrum** X^V of V as the suspension spectrum of the Thom space $\mathrm{Th}(V)$ of V:

$$X^V \coloneqq \Sigma^\infty \mathrm{Th}(V).$$

We extend this Thom spectrum construction to virtual vector bundles as follows. If X is compact (e.g., a finite complex), every virtual vector bundle $\alpha = E_1 - E_2$ on X can be expressed as the difference $E - \boldsymbol{n}_X$ of a genuine vector bundle E and a trivial vector bundle of sufficiently high rank. That is, we have an isomorphism of vector bundles $E_1 \oplus \boldsymbol{n}_X \cong E \oplus E_2$. We define the *Thom spectrum* X^α of α by

$$X^\alpha \coloneqq \Sigma^{-n} X^E.$$

We can also extend other operations to virtual vector bundles.

Definition 1.11. Given a continuous map $f : Y \to X$, the **pullback** $f^*(\alpha)$ of α along f is the virtual vector bundle over Y defined by

$$f^*(\alpha) \coloneqq f^*(E_1) - f^*(E_2).$$

1.3.1. *Madsen-Tillmann Spectra.* Let $\boldsymbol{K} = O$ or SO.

Definition 1.12. Given $n \geq 0$, the **Madsen-Tillmann spectrum** $MT\boldsymbol{K}(n)$ is defined as the Thom spectrum of the virtual vector bundle $-\gamma_n^{\boldsymbol{K}}$ over $B\boldsymbol{K}(n)$.

Let $M \to E \xrightarrow{f} B$ be a smooth fiber bundle such that M is a closed manifold of dimension n with $\boldsymbol{K}(n)$-structure. In particular f is a surjective submersion, hence we have a short exact sequence of vector bundles on E

$$0 \longrightarrow T_f E \longrightarrow TE \longrightarrow f^*TB \longrightarrow 0$$

where the kernel $T_f E$ is called the *vertical or (fiberwise) tangent bundle* of f. Note $T_f E$ naturally inherits a $\boldsymbol{K}(n)$-structure, hence it is classified by a map $\psi : E \to B\boldsymbol{K}(n)$, unique up to homotopy. This yields a map at the level of Thom spectra:

$$\kappa_E : E^{-T_f E} \to MT\boldsymbol{K}(n).$$

Fix a smooth embedding $i : E \hookrightarrow \mathbf{R}^n$. From this, we define a fiberwise inclusion $j = (f, i) : E \to \boldsymbol{n}_B$ with normal bundle ν_j. We have the following short exact sequence of vector bundles on E:

$$0 \longrightarrow TE \longrightarrow f^*TB \oplus \boldsymbol{n}_E \longrightarrow \nu_j \longrightarrow 0\ .$$

In particular, combining the two short exact sequence above yields the short exact sequence

$$0 \longrightarrow T_f E \longrightarrow \boldsymbol{n}_E \longrightarrow \nu_j \longrightarrow 0$$

hence we get an identification of virtual vector bundles $-T_f E = \nu_j - \boldsymbol{n}_E$. The Pontryagin-Thom collapse applied to j yields a map of Thom spectra

$$f^! : \Sigma_+^{\infty+n} B \to \Sigma^n E^{-T_f E}.$$

Desuspending gives us a map $f^! : \Sigma_+^\infty B \to E^{-T_f E}$.

Definition 1.13. We define the **Madsen-Tillmann-Weiss map** (or **MTW map**) ω_E of the bundle f as the composite

$$\omega_E : \Sigma_+^\infty B \xrightarrow{f^!} E^{-T_f E} \xrightarrow{\kappa_E} MT\boldsymbol{K}(n)\ .$$

Example 1.14. The canonical embedding $i : \boldsymbol{K}(n) \hookrightarrow \boldsymbol{K}(n+1)$ induces, via Proposition 1.5, an n-sphere bundle $S^n \longrightarrow B\boldsymbol{K}(n) \xrightarrow{Bi} B\boldsymbol{K}(n+1)$. We denote by ω the MTW map

$$\omega : \Sigma^\infty_+ B\boldsymbol{K}(n+1) \to MT\boldsymbol{K}(n)$$

associated to Bi.

There are two more maps involving Madsen-Tillmann spectra we would like to define. With our Grassmannian model for $B\boldsymbol{K}(n)$, $n \geq 1$, the natural inclusion $\mathbf{R}^k \to \mathbf{R}^{k+1}$ of the first k-coordinates induces an embedding $\mathrm{Gr}_n(\mathbf{R}^k) \hookrightarrow \mathrm{Gr}_n(\mathbf{R}^{k+1})$ (and similarly in the oriented case). Passing to the colimit as $k \to \infty$ gives an embedding $j : B\boldsymbol{K}(n) \to B\boldsymbol{K}(n+1)$ as well as a bundle isomorphism $U^{\boldsymbol{K}}_{n+1}|_{B\boldsymbol{K}(n)} \cong U^{\boldsymbol{K}}_n \oplus \mathbf{1}$. That is, we have a pullback square

$$\begin{tikzcd} U^{\boldsymbol{K}}_n \oplus \mathbf{1} \arrow[r] \arrow[d, "\gamma^{\boldsymbol{K}}_n \oplus \mathbf{1}"'] \arrow[dr, phantom, "\lrcorner"] & U^{\boldsymbol{K}}_{n+1} \arrow[d, "\gamma^{\boldsymbol{K}}_{n+1}"] \\ B\boldsymbol{K}(n) \arrow[r, "j"'] & B\boldsymbol{K}(n+1) \end{tikzcd}$$

Passing to Thom spectra induces a map

$$\eta : MT\boldsymbol{K}(n) \to \Sigma MT\boldsymbol{K}(n+1).$$

Note η is the virtual analog of the nth structure map of the Thom spectrum $M\boldsymbol{K}$.

Given $n \geq 1$, since $B\boldsymbol{K}(n)$ is a CW-complex, $U^{\boldsymbol{K}}_n$ is a direct summand of a trivial bundle $U^{\boldsymbol{K}}_n \oplus E \cong \boldsymbol{m}_{B\boldsymbol{K}(n)}$, hence $MT\boldsymbol{K}(n) \simeq \Sigma^{-m} B\boldsymbol{K}(n)^E$. The bundle inclusion $E \hookrightarrow \boldsymbol{m}_{B\boldsymbol{K}(n)}$ induces a map

$$\beta : MT\boldsymbol{K}(n) \to \Sigma^\infty_+ B\boldsymbol{K}(n),$$

unique up to homotopy.

The following is Proposition 3.1 of [GMTW09].

Proposition 1.15. *We have a cofiber sequence*

$$MT\boldsymbol{K}(n+1) \xrightarrow{\beta} \Sigma^\infty_+ B\boldsymbol{K}(n+1) \xrightarrow{\omega} MT\boldsymbol{K}(n) \xrightarrow{\eta} \Sigma MT\boldsymbol{K}(n+1) \ .$$

1.4. **Localizations and splittings of spectra.**

Throughout we will work within the stable homotopy category $\mathrm{Ho}(\mathsf{Sp}_{(p)})$ localized at various primes p. Formally speaking, this category is constructed by first taking the Bousfield localization $\mathsf{Sp}_{(p)} \coloneqq L_{M(\mathbf{Z}_{(p)})}\mathsf{Sp}$ of the category of spectra with respect to smashing with the Moore spectrum $M(\mathbf{Z}_{(p)})$. A detailed version of this construction can be found in Chapter 7 of [BR20]. For our purposes we will simply state the desirable properties of the p-localized category of spectra that we will need.

Proposition 1.16. *There exists a spectrum $M(\mathbf{Z}_p)$ such that smashing with $M(\mathbf{Z}_p)$ defines an endofunctor $(-)_{(p)} \coloneqq - \wedge M(\mathbf{Z}_{(p)})$ on the category of spectra such that, given a spectrum E, we have an isomorphism of graded abelian groups*

$$\pi_*(X_{(p)}) \cong \pi_*(X)_{(p)}.$$

Corollary 1.17.

1. *Given an endomorphism $f : \mathbf{S} \to \mathbf{S}$, $f_{(p)}$ is a homotopy equivalence iff $p \nmid \deg(f)$.*
2. *Given a spectrum E and an abelian group A, we have an isomorphism of graded abelian groups*

$$H^*(E_{(p)}; A) \cong \pi_{-*}F(E_p, HA_{(p)}) \simeq \pi_{-*}F(E, HA_{(p)}) \cong H^*(E; \mathbf{Z}) \otimes_{\mathbf{Z}} A_{(p)}$$

and similarly for homology.

We now fix some terminology related to splitting spectra. As the stable homotopy category is triangulated, the language reflects the situation in a general additive category, such as the familiar case of modules over a commutative ring.

Definition 1.18. A **splitting** of a spectrum E is a homotopy equivalence $f : E_1 \vee E_2 \xrightarrow{\simeq} E$. Given such a splitting f, we say E **splits** and E_1 and E_2 **split off** E.

In practice, one often begins with a map of spectra $i : E_1 \to E$, and asks whether E_1 splits off E. As in the case of modules, a splitting of a spectrum coincides with a split short exact sequence.

Lemma 1.19.

1. *Given a map of spectra $s : E_1 \to E$, E_1 splits off E if and only if there exists a map $r : E \to E_1$ such that rs is a self-homotopy equivalence.*
2. *Given a map of spectra $r : E \to E_2$, E_2 splits off E if and only if there exists a map $s : E_2 \to E$ such that rs is a self-homotopy equivalence.*

Proof. In the stable homotopy category this amounts to a routine verification that in an additive category, a splitting of an object coincides with a split short exact sequence of its summands. In the first case, we have $E \simeq E_1 \vee \mathrm{hocofib}(s)$. In the second case, we have $E \simeq E_2 \vee \mathrm{hofib}(r)$. □

2. Transfers

2.1. (Twisted) Umkehr maps.

Let $f : M \hookrightarrow N$ be an embedding of closed manifolds. The Pontryagin-Thom collapse yields a "wrong way" based map

$$f^! : N^+ \to M^{\nu_f}$$

where N^+ denotes the one-point compactification of n (with basepoint at infinity) and ν_f is the normal bundle of f. We call $f^!$ the *Umkehr map* associated to f. Such maps are ubiquitous throughout classical algebraic and differential topology. Given a smooth vector bundle ξ over N, we also have a notion of a *twisted umkehr map* between Thom spectra

$$f^!_\xi : N^\xi \longrightarrow M^{\nu_f \oplus f^*\xi}.$$

This is done by applying the Pontryagin-Thom collapse to the composite

$$f' : M \xrightarrow{f} N \xrightarrow{0} D(\xi) \ .$$

Explicitly, denoting by $\pi : E \to N$ the projection of ξ, as π is a surjective submersion, as before we get a short exact sequence of vector bundles over E:

$$0 \longrightarrow T_\pi E \longrightarrow TE \longrightarrow \pi^* TN \longrightarrow 0 \ .$$

Pulling back along the zero section $0 : N \hookrightarrow D(\xi)$ yields a short exact sequence of vector bundles over N:

$$0 \longrightarrow E \longrightarrow TE|_N \longrightarrow TN \longrightarrow 0 \ .$$

As N is compact, we can endow $TE|_N$ with a Riemannian metric, hence we get a splitting $TE|_M \cong E \oplus TN$. This induces a splitting of normal bundles

$$\nu_{f'} \cong \nu_f \oplus f^* E.$$

Finally, since the base M is compact, the Thom space of ξ coincides with the one-point compactification of $D(\xi)$, hence the Pontryagin-Thom collapse yields the desired twisted umkehr map $f^!_\xi$.

We introduce three particularly important examples of twisted umkehr maps, called transfers.

2.2. The Boardman transfer.

Let $f : M \to N$ be an arbitrary smooth map between closed manifolds. By the Whitney embedding theorem, there exists an embedding $i : M \hookrightarrow \mathbf{R}^n$ for sufficiently large n. Define the embedding $j : M \hookrightarrow \boldsymbol{n}_N = N \times \mathbf{R}^n$ by $j = (f, i)$, and let ν_j be the associated normal bundle. The Pontryagin-Thom collapse defines an umkehr map

$$f^! : \Sigma^n N_+ \cong \mathbf{n}_N^+ \longrightarrow \mathrm{Th}(\nu_j).$$

Given a vector bundle ξ over N, we can obtain a twisted umkehr map by applying the Pontryagin-Thom collapse to the embedding given by the composite

$$j' : M \xrightarrow{j} \boldsymbol{n}_N \longrightarrow \xi \oplus \boldsymbol{n}_N \ .$$

A choice of Riemannian metric on $f^*(\xi \oplus \boldsymbol{n}_N) \cong f^*(\xi) \oplus \boldsymbol{n}_M$ yields a splitting of normal bundles $\nu_{j'} \cong f^*(\xi) \oplus \nu_j$. From this we obtain a twisted umkehr map

$$f^! : \Sigma^n \mathrm{Th}(\xi) \cong \mathrm{Th}(\xi \oplus \boldsymbol{n}_N) \longrightarrow \mathrm{Th}(f^*\xi \oplus \nu_j).$$

While this construction depends on the choice n, passing to the stable homotopy category we obtain a well-defined stable map of Thom spectra

$$\widetilde{t}^\xi_f : \Sigma^n N^\xi \to M^{f^*\xi \oplus \nu_j}$$

which we call the *Boardman transfer* of f *twisted by* ξ.

2.3. **The Becker-Gottlieb transfer.** Let $F \to E \xrightarrow{f} B$ be a fiber bundle with F, E, B compact manifolds and $j : E \hookrightarrow \boldsymbol{n}_B$ be a fiberwise embedding over B. We have the short exact sequence of vector bundles on E:

$$0 \longrightarrow TE \longrightarrow f^*(TB) \oplus \boldsymbol{n}_E \cong j^*(T\boldsymbol{n}_B) \longrightarrow \nu_j \longrightarrow 0 \ .$$

As in the definition of the MTW map ω_f, since f is a surjective submersion, we also have a natural exact sequence of vector bundles

$$0 \longrightarrow T_f E \longrightarrow TE \longrightarrow f^*(TB) \longrightarrow 0 \ .$$

Endowing the vector bundles over E with metrics gives splittings $TE \oplus \nu_j \cong f^*(TB) \oplus \boldsymbol{n}_E$ and $TE \cong T_f E \oplus f^*(TB)$, and combining the two yields the trivialization

$$T_f E \oplus \nu_j \cong \boldsymbol{n}_E.$$

Thus we have an inclusion of bundles $\nu_j \to \boldsymbol{n}_E$, yielding an associated map of Thom spectra $E^{\nu_j} \to \Sigma_+^{\infty+n} E$, unique up to homotopy. Precomposing this with the (untwisted) Boardman transfer of f yields a map of Thom spectra

$$t_f : \Sigma_+^{\infty+n} B \xrightarrow{\widetilde{t}_f} E^{\nu_j} \longrightarrow \Sigma_+^{\infty+n} E.$$

called the *Becker-Gottlieb transfer* of f. As in the case of the Boardman transfer, given a vector bundle ξ on the base B, we can obtain a ξ-twisted variant of the Becker-Gottlieb transfer.

2.4. **The Becker-Schultz-Mann-Miller-Miller-transfer.** Our major splitting results depend on yet another variant of the Boardman transfer, defined in Section 2 of [MMM86], which we call the *BSMMM transfer*.

Throughout we fix a compact Lie group G with associated Lie algebra $\mathfrak{g}$. Recall $\mathfrak{g}$ admits a (left) G-action via the adjoint representation Ad_G.

Definition 2.1. Let M be a manifold with a right G-action which is free and smooth. We define the **adjoint vector bundle** $\mathrm{ad}_{G,M}$ over M/G as the vector bundle with fiber $\mathfrak{g}$ associated to the principal G-bundle M:

$$\mathrm{ad}_{G,M} \coloneqq M \times_G \mathfrak{g}.$$

In the case $M = EG$, we write $\mathrm{ad}_G \coloneqq \mathrm{ad}_{G,EG}$.

Suppose M is compact. It follows that M/G is compact, hence we can endow TM/G with a metric and obtain a splitting of vector bundles over M/G ([MMM86] Lemma 2.1):

$$TM/G \cong \mathrm{ad}_{G,M} \oplus T(M/G),$$

where the action of G on the tangent bundle TM is given by the differential $g \mapsto dg$. Furthermore, this splitting is natural with respect to the G-manifold M.

Given a closed subgroup $K < G$, we have a smooth fiber bundle $G/K \to M/K \xrightarrow{f} M/G$. As before, we can choose a fiberwise embedding $j' = (j, f) : M/K \hookrightarrow \boldsymbol{n}_{M/G}$, and define the associated Boardman transfer of f twisted by $\mathrm{ad}_{G,M}$:

$$\widetilde{t}_f^{\mathrm{ad}_{G,M}} : \Sigma^n (M/G)^{\mathrm{ad}_{G/M}} \longrightarrow (M/K)^{\nu_j \oplus f^* \mathrm{ad}_{G,M}}.$$

In this particular case, we can identify the target Thom spectrum with a particularly simple model.

Firstly, we have a commutative diagram of vector bundles over M/K with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{ad}_{K,M} & \longrightarrow & TM/K & \longrightarrow & T(M/K) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow\cong & & \downarrow & & \\ 0 & \longrightarrow & f^* \mathrm{ad}_{G,M} & \longrightarrow & f^*(TM/G) & \longrightarrow & f^*T(M/G) & \longrightarrow & 0 \end{array}$$

Since the middle map is a bundle isomorphism, the snake lemma implies we can naturally identify the vertical tangent bundle of f with the cokernel of the left vertical map:

$$0 \longrightarrow \mathrm{ad}_{K,M} \longrightarrow f^* \mathrm{ad}_{G,M} \longrightarrow T_f E \longrightarrow 0 \ .$$

This data is called a *relative trivialization* of $f, \mathrm{ad}_{K,M}, f^* \mathrm{ad}_{G,M}$. In particular, splitting the above sequence induces a bundle isomorphism $\nu_j \oplus f^* \mathrm{ad}_{G,M} \cong \nu_j \oplus (T_f E \oplus \mathrm{ad}_{K,M}) \cong \boldsymbol{n}_E \oplus \mathrm{ad}_{K,M}$. At the level of Thom spectra, this isomorphism implies we can rewrite the Boardman transfer as

$$\widetilde{t}_f^{\mathrm{ad}_{G,M}} : (M/G)^{\mathrm{ad}_{G,M}} \longrightarrow (M/K)^{\mathrm{ad}_{K,M}}.$$

In the case $M = EG$, we can filter EG by compact free G-manifolds M_α, such that ad_G restricts to ad_{G,M_α} over each M_α, hence we get a well-defined transfer

$$\widetilde{t}_{K,G} : BG^{\mathrm{ad}_G} \longrightarrow BK^{\mathrm{ad}_K}$$

called the *untwisted BSMMM transfer* associated to K, G. Unsurprisingly, as the BSMMM transfer coincides with the Boardman transfer up to an automorphism of the source Thom spectrum (which depends on the homotopy type of the choice of relative trivialization), we can twist the BSMMM transfer by a vector bundle ξ on BG.

2.4.1. *Relating the Becker-Gottlieb and BSMMM transfers.* As constructed above, the BSMMM transfer, up to a homotopy automorphism of the target, coincides with the Boardman transfer. It follows that in the case of a compact Lie group G with a closed subgroup $K < G$, the Becker-Gottlieb transfer t_f of the fiber bundle $G/K \to BK \xrightarrow{f} BG$ should also factor through the BSMMM transfer $\widetilde{t}_{K,G}$, and in fact this is the case.

Proposition 2.2. *The Becker-Gottlieb transfer t_f admits a factorization along the BSMMM transfer of f:*

$$t_f : \Sigma^\infty_+ BG \xrightarrow{\widetilde{t}_{K,G}} BK^{\mathrm{ad}_K - f^* \mathrm{ad}_G} \longrightarrow \Sigma^\infty_+ BK \ .$$

The same is true for the Becker-Gottlieb transfer of f twisted by a (possibly virtual) vector bundle ξ over BG:

$$t_f^\xi : BG^\xi \xrightarrow{\widetilde{t}^\xi_{K,G}} BK^{f^*\xi \oplus \mathrm{ad}_K - f^* \mathrm{ad}_G} \longrightarrow BK^{f^*\xi} \ .$$

3. Splitting Madsen-Tillman Spectra

3.1. A splitting via the Becker-Gottlieb transfer.

Our first splitting results for Madsen-Tillman spectra rely on a spectra-level generalization of Hopf's vector field theorem by Becker and Gottlieb. Namely, they showed that for a smooth fiber bundle $f : E \to B$ with compact fiber F, the Becker-Gottlieb transfer t_f encodes the Euler characteristic of the fiber.

Theorem 3.1 ([BG74], Theorem 5.5)**.** *Given an abelian group A, the composite*

$$\Sigma^\infty_+ B \xrightarrow{t_f} \Sigma^\infty_+ E \xrightarrow{\Sigma^\infty_+ f} \Sigma^\infty_+ B$$

induces multiplication by $\chi(F)$ on $H^*(-; A)$-cohomology.

In particular, if p does not divide $\chi(F)$, the above composite is a self-homotopy equivalence in the p-localized category of spectra.

Corollary 3.2. *If $\chi(F)$ is prime to p, $(\Sigma_+ B)_{(p)}$ splits off $(\Sigma^\infty_+ E)_{(p)}$*

Let $\boldsymbol{K} = O$ or SO. Theorem 3.1 gives a criterion for when the sphere spectrum $\mathbf{S}$ splits off the Madsen-Tillman spectra $MT\boldsymbol{K}(n)$, in terms of the existence of a manifold with $\boldsymbol{K}(n)$-structure.

Theorem 3.3. *Suppose there exists a manifold M with $\boldsymbol{K}(n)$-structure. If $\chi(M)$ is prime to p, $\mathbf{S}^0_{(p)}$ splits off $MT\boldsymbol{K}(n)_{(p)}$. In particular,*

1. *$\mathbf{S}^0$ splits off $MTO(2n)$.*
2. *For p odd, $\mathbf{S}^0_{(p)}$ splits off $MTSO(2n)$.*

Proof. We first prove the general splitting result. Suppose M is an n-dimensional manifold with structure group $\boldsymbol{K}(n)$. Let $j : M \hookrightarrow \mathbf{R}^m$ be a smooth embedding. The vertical tangent bundle associated to the trivial fiber bundle $c : M \to *$ is TM, hence by definition the MTW map ω_M of M (as a bundle) is the composite

$$\omega_M : \mathbf{S}^0 \xrightarrow{\widetilde{t}_M} M^{-TM} \xrightarrow{\kappa_M} MT\boldsymbol{K}(n)$$

where the first map is the Boardman transfer of M, and κ_M is induced by functoriality of the Thom spectrum applied to the classifying map $f : M \to B\boldsymbol{K}(n)$ of TM. Explicitly, viewing j as being a fiberwise embedding over the point, a choice of splitting $\boldsymbol{m}_E \cong TM \oplus \nu_j$ via the natural metric on the trivial vector bundle gives bundle inclusion and projection maps $i : \nu_j \hookrightarrow \boldsymbol{m}_E$ and $p : \boldsymbol{m}_E \hookrightarrow \nu_j$. The maps p and pi induce the Boardman transfer $\widetilde{t}_M$ and Becker-Gottlieb transfer t_M of M, respectively. Consequently, we have the following commutative diagram of spectra

$$\begin{tikzcd}
\mathbf{S}^0 \arrow[r, "\widetilde{t}_M"] \arrow[d, equal] & M^{-TM} \arrow[r] \arrow[d, "\kappa_M"] & \Sigma^\infty_+ M \arrow[r, "\Sigma^\infty_+ c"] \arrow[d, "\Sigma^\infty_+ f"] & \mathbf{S}^0 \arrow[d, equal] \\
\mathbf{S}^0 \arrow[r, "\omega_{\boldsymbol{K}}"'] & MT\boldsymbol{K}(n) \arrow[r, "\beta_{\boldsymbol{K}}"'] & \Sigma^\infty_+ B\boldsymbol{K}(n) \arrow[r, "\Sigma^\infty_+ c"'] & \mathbf{S}^0
\end{tikzcd} .$$

In particular, Theorem 3.1 implies the composite given by the top row induces multiplication by $\chi(M)$ on singular cohomology, hence so does the composite given by the bottom row. That is, the bottom composite is a map of degree $\chi(M)$, hence for p prime to $\chi(M)$, we get that $\boldsymbol{S}^0_{(p)}$ splits off $MT\boldsymbol{K}(n)_{(p)}$ as required.

For the two remaining claims, the first splitting follows from the fact that $\mathbf{RP}^{2n}$ is a smooth $2n$-dimensional manifold with $\chi(\mathbf{RP}^{2n}) = 1$, while the second follows from the fact that S^{2n} is orientable with Euler characteristic 2. □

3.2. **Splittings via the twisted Becker-Gottlieb transfer.** In order to establish further splittings of Madsen-Tillmann spectra, we need the following twisted analog of Theorem 3.1.

Theorem 3.4. *Let $F \to E \xrightarrow{f} B$ be a fiber bundle with F, E, B compact manifolds and ξ a vector bundle over B. The composite map of Thom spectra*

$$B^\xi \xrightarrow{t^\xi_f} E^{f^*\xi} \xrightarrow{\mathrm{Th}(f)} B^\xi ,$$

where the first map is the Becker-Gottlieb transfer of f twisted by ξ, and the second map is induced by functoriality of the Thom space, induces multiplication by $\chi(F)$ on $H^*(-;A)$ for any abelian group A. In particular, if $\chi(F)$ is prime to p, B^ξ splits off $E^{f^*\xi}$.

Our main application of Theorem 3.4 will be for the Becker-Gottlieb transfer t_f of the fiber bundle $G/K \to BK \xrightarrow{f} BG$ for G a Lie group and $K < G$ a closed subgroup. In such a case, the following is a simple consequence of the fact that t_f factors along the BSMMM transfer $\widetilde{t}_{K,G}$ (Proposition 2.2).

Corollary 3.5. *Suppose p is a prime which does not divide $\chi(G/K)$. Given a vector bundle ξ over BG, $BG^\xi_{(p)}$ splits off $BK^{f^*\xi \oplus \mathrm{ad}_K - f^*\,\mathrm{ad}_G}_{(p)}$. Taking ξ to be the trivial bundle of rank 0 implies $\Sigma^\infty_+ BG_{(p)}$ splits off $BK^{\mathrm{ad}_K - f^*\,\mathrm{ad}_G}_{(p)}$.*

3.3. **Identifying the cofiber of the transfers.**

Corollary 3.5 along with Lemma 1.19 tell us that for a closed subgroup K of a compact Lie group G and a prime $p \not| \chi(G/K)$, we get a splitting of Thom spectra

$$BK^{f^*\xi \oplus \mathrm{ad}_K - f^*\,\mathrm{ad}_G}_{(p)} \simeq BG^\xi_{(p)} \vee \mathrm{hocofib}(\widetilde{t}^\xi_{K,G})_{(p)},$$

where $\widetilde{t} := \widetilde{t}^\xi_{K,G}$ is the BSMMM transfer of (K, G) twisted by a vector bundle $\xi \to BG$.

The following Theorem due to Morisugi [Mor09] gives us suitable conditions under which we can identify this cofiber.

Theorem 3.6. *Let G be a compact Lie group and $K < G$ a closed subgroup such that there exists an orthogonal G-representation V with $G/K \cong S(V)$ as G-spaces. Denote by λ the associated vector bundle $EG \times_G V$ over BG. Given a (possibly virtual) vector bundle ξ over BG, we have a cofiber sequence of Thom spectra*

$$BG^{\mathrm{ad}_G \oplus \xi - \lambda} \longrightarrow BG^{\mathrm{ad}_G \oplus \xi} \xrightarrow{\widetilde{t}} BK^{\mathrm{ad}_K \oplus \xi|_K} \longrightarrow \Sigma BG^{\mathrm{ad}_G \oplus \xi - \lambda} .$$

We can apply this above theorem to get the cofiber sequence of Proposition 1.15, substituting the MTW map ω with the twisted BSMMM transfer.

Lemma 3.7. *Let $\boldsymbol{K} = O$ or SO. We have a cofiber sequence of Thom spectra*

$$MT\boldsymbol{K}(n+1) \xrightarrow{\beta} \Sigma^\infty_+ B\boldsymbol{K}(n+1) \xrightarrow{\widetilde{t}} MT\boldsymbol{K}(n) \xrightarrow{\eta} \Sigma MT\boldsymbol{K}(n+1) \ .$$

Proof. The Lie group $\boldsymbol{K}(n+1)$ admits a canonical representation on $\mathbf{R}^{n+1}$ which we denote by $\gamma^{\boldsymbol{K}}_{n+1}$.[1] The $\boldsymbol{K}(n)$-action on $\mathbf{R}^{n+1}$ restricts to a transitive action on the unit sphere $S(\mathbf{F}^{n+1})$ of $\mathbf{R}^n$. The stabilizer of $\boldsymbol{K}(n+1)$ with respect to a vector $v \in S(\mathbf{R}^{n+1})$ is $\boldsymbol{K}(n) \subset \boldsymbol{K}(n+1)$, hence by the orbit-stabilizer theorem for Lie groups, we have a diffeomorphism $\boldsymbol{K}(n+1)/\boldsymbol{K}(n) \cong S(\mathbf{R}^{n+1})$. We apply Theorem 3.7, taking $G = \boldsymbol{K}(n+1)$, $K = \boldsymbol{K}(n)$, $V = \mathbf{R}^{n+1}$, with twisting $\xi = -\operatorname{ad}_{\boldsymbol{K}(n+1)}$ over $B\boldsymbol{K}(n+1)$. This gives the cofiber sequence

$$MT\boldsymbol{K}(n+1) \xrightarrow{\beta} B\boldsymbol{K}(n+1)^{\mathbf{0}} \cong \Sigma^\infty_+ B\boldsymbol{K}(n+1) \xrightarrow{\widetilde{t}^{\xi}_{K,G}} BK^{\operatorname{ad}_{\boldsymbol{K}(n)} - \operatorname{ad}_{\boldsymbol{K}(n+1)}|_{\boldsymbol{K}(n)}} \longrightarrow \Sigma MT\boldsymbol{K}(n+1) \ .$$

It remains to identify the pullback bundle $\operatorname{ad}_{\boldsymbol{K}}(n+1)|_{\boldsymbol{K}(n)}$ over $\boldsymbol{K}(n)$. We prove the case $\boldsymbol{K} = O$ for clarity; the cases $\boldsymbol{K} = SO, U$ are proven analogously. We have a natural inclusion $i : O(n) \hookrightarrow O(n+1)$ given by

$$i : A \mapsto \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

By Proposition 1.5, we can model Bi as the projection map $EO(n+1)/O(n) \to EO(n+1)/O(n+1)$. We get the following isomorphisms of bundles

$$\begin{aligned} \operatorname{ad}_{O(n+1)}|_{O(n)} &= (Bi)^*(\operatorname{ad}_{O(n+1)}) \\ &= (Bi)^*(EO(n+1) \times_{O(n+1)} \mathfrak{o}(n+1)) \\ &\cong ((Bi)^* EO(n+1)) \times_{O(n+1)} \mathfrak{o}(n+1) \\ &\cong (EO(n+1) \times_{O(n)} O(n+1)) \times_{O(n+1)} \mathfrak{o}(n+1) \\ &\cong EO(n+1) \times_{O(n)} \mathfrak{o}(n+1), \end{aligned}$$

where the left-action of $O(n)$ on $\mathfrak{o}(n)$ is given by restricting the adjoint action by $O(n+1)$.

In particular, given $A \in \mathfrak{o}(n+1)$, we have

$$A = \begin{pmatrix} A' & B \\ -B^* & 0 \end{pmatrix}$$

where $A' \in \mathfrak{o}(n)$ and $B \in M_{n,1}(\mathbf{R}) \cong \mathbf{R}^n$. Thus given $X \in O(n) \subset O(n+1)$, the adjoint action is given by

$$XA = \begin{pmatrix} X & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} A' & B \\ -B^* & 0 \end{pmatrix} \begin{pmatrix} X^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} XA'X^{-1} & XB \\ -(XB)^* & 0 \end{pmatrix}.$$

Thus we have a decomposition of $O(n)$-representations $\mathfrak{o}(n+1) \cong \mathfrak{o}(n) \oplus \mathbf{R}^n$ and consequently a splitting of vector bundles $\operatorname{ad}_{O(n+1)}|_{O(n)} \cong \operatorname{ad}_{O(n)} \oplus \gamma^{\boldsymbol{K}}_n$. This implies the above cofiber sequence is as desired, completing the proof. □

3.4. **Main splitting results.**

Theorem 3.8.

1. *$\Sigma^\infty_+ BSO(2n+1)$ splits off $MTO(2n)$ with no localization involved.*
2. *Let p be an odd prime with $\boldsymbol{K} = O, SO$. We have a splitting*

$$MTK(2n)_{(p)} \simeq \Sigma^\infty_+ BK(2n+1)_{(p)} \vee \Sigma MTK(2n+1)_{(p)}.$$

Proof. 1. We prove the first splitting by applying Corollary 3.5 to the embedding $j : O(2n) \hookrightarrow SO(2n+1)$ defined by

$$j : A \mapsto \det(A) \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

Note j lands in $SO(2n+1)$ as given $A \in O(2n)$, $\det(j(A)) = \det(\det(A)A)\det(A) = \det(A)^{2(n+1)} = 1$.

By Proposition 1.5, we have a fibration sequence

$$SO(2n+1)/O(2n) \longrightarrow BSO(2n+1)/O(2n) \simeq BO(2n) \xrightarrow{Bj} BSO(2n+1) \ .$$

[1] Strictly speaking, this is an abuse of notation as it coincides with the tautological n-plane bundle. We justify it by the fact that the associated vector bundle $E\boldsymbol{K}(n+1) \times_{\boldsymbol{K}(n)} \gamma^{\boldsymbol{K}}_n$ is the tautological n-plane bundle over $B\boldsymbol{K}(n)$.

The canonical action of $SO(2n+1)$ on $\mathbf{R}^{2n+1}$ descends to a transitive action on $\mathbf{R}P^{2n}$ with stabilizer $O(2n) \subset SO(2n+1)$, hence via the orbit-stabilizer theorem for compact Lie groups we have a diffeomorphism $SO(2n+1)/O(2n) \cong \mathbf{R}P^{2n}$, with $\chi(\mathbf{R}P^{2n}) = 1$. It remains to identify the pullback bundle $j^*(\mathrm{ad}_{SO(2n+1)})$ over $BO(2n)$.

As in the proof of Lemma 3.7, we can identify the pullback bundle with the associated vector bundle with fiber given by the Lie algebra of $SO(n+1)$:

$$(Bj)^*(\mathrm{ad}_{SO}(2n+1)) \cong ESO(2n+1) \times_{O(2n)} \mathfrak{so}(2n+1).$$

A matrix $A \in \mathfrak{so}(2n+1)$ can be written as

$$A = \begin{pmatrix} A' & B \\ -B^* & 0 \end{pmatrix}$$

with $A' \in \mathfrak{so}(2n) = \mathfrak{o}(2n)$ and $B \in M_{2n,1}(\mathbf{R}) \cong \mathbf{R}^{2n}$. Thus given $X \in O(2n) \subset O(2n+1)$, the adjoint action is given by

$$X \cdot A = \begin{pmatrix} \det(X)X & 0 \\ 0 & \det(X) \end{pmatrix} \begin{pmatrix} A' & B \\ -B^* & 0 \end{pmatrix} \begin{pmatrix} \det(X)^{-1}X^{-1} & 0 \\ 0 & \det(X)^{-1} \end{pmatrix} = \begin{pmatrix} XA'X^{-1} & XB \\ -(XB)^* & 0 \end{pmatrix}.$$

This gives an isomorphism of $O(2n)$-representations $\mathfrak{so}(2n+1) \cong \mathfrak{o}(2n) \oplus \mathbf{R}^{2n}$, hence we get a splitting of vector bundles $(Bj)^*(\mathrm{ad}_{SO(2n+1)}) \cong \mathrm{ad}_{O(2n)} \oplus \gamma^O_{2n}$. Taking ξ to be the trivial bundle of rank 0 over $BSO(2n+1)$ and Bj to be the fiber bundle f in Corollary 3.5 gives us the desired splitting.

2. For the second splitting result, denote by j the standard embedding $O(2n) \subset O(2n+1)$ as in the proof of Lemma 3.7. It follows that Bj defines a fiber sequence with fiber $S^{2n} \cong O(2n+1)/O(2n)$. Restricting j to the connected component of the identity yields a fiber bundle $BSO(2n) \to BSO(2n+1)$, again with fiber $S^{2n} \cong SO(2n+1)/SO(2n)$. Since $\chi(S^{2n}) = 2$ is prime to p, again by Corollary 3.5, the p-localized BSMMM transfer $\widetilde{t} : \Sigma^\infty_+ B\boldsymbol{K}(2n+1)_{(p)} \to B\boldsymbol{K}(2n)_{(p)}^{\mathrm{ad}_{\boldsymbol{K}(2n)} - \mathrm{ad}_{\boldsymbol{K}(2n+1)}|_{\boldsymbol{K}(2n)}}$ provides a splitting

$$\Sigma MT\boldsymbol{K}(2n) \simeq B\boldsymbol{K}(2n)_{(p)}^{\mathrm{ad}_{\boldsymbol{K}(2n)} - \mathrm{ad}_{\boldsymbol{K}(2n+1)}|_{\boldsymbol{K}(2n)}} \simeq \Sigma^\infty_+ B\boldsymbol{K}(2n+1)_{(p)} \vee \mathrm{hocofib}(\widetilde{t})_{(p)}.$$

By Lemma 3.7, we can identify the cofiber of $\widetilde{t}$ to obtain the desired splitting. □

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